
ALGEBRA-SUMMER

Vincent Ferrara

Contents

0.1	Intro	3
1	Groups	5
1.1	Intro	5
1.1.1	Exercises	5
1.2	Dihedral Groups	6
1.2.1	Symmetry Group for an Equilateral Triangle	6
1.2.2	Symmetry Group for a Square	8
1.2.3	Symmetry Group of regular n-gon	9
1.2.4	Exercises	10
2	Symmetric Groups	11
2.1	Cycle Notation	12

0.1 Intro

Theorem 0.1.1. (Division Algorithm) Suppose $a, b \in \mathbb{Z}$ s.t. $b > 0$. Then there are unique $q, r \in \mathbb{Z}$ s.t. $a = bq + r$ and $0 \leq r < b$

Proof. Let $a, b \in \mathbb{Z}$ s.t. $b > 0$. Consider $A = \{a - bx \mid x \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this r and call the associated $x \in \mathbb{Z}$, q .

Thus, $r = a - bq \implies a = bq + r$.

For contradiction, assume $r \geq b$.

$\implies r = b + m$ s.t. $m \geq 0$

$\implies b + m = a - bq \implies m = a - b(q + 1) \in A \implies m < r = \text{minimum} \implies \Leftarrow$ Thus $0 \leq r < b$

Fix $q_1, q_2, r_1, r_2 \in \mathbb{Z}$ s.t. $a = bq_1 + r_1 = bq_2 + r_2$ and $0 \leq r_1, r_2 < b$. WLOG let $r_1 \geq r_2$

$\implies bq_1 + r_1 = bq_2 + r_2 \implies r_1 - r_2 = b(q_2 - q_1) \implies r_1 - r_2$ is a multiple of b

Since $0 \leq r_1, r_2 < b \implies 0 \leq r_1 - r_2 < b$

Since $r_1 - r_2$ is strictly less than b the only multiple of b it could be is 0 thus $r_1 = r_2$

$\implies q_1 = q_2$ □

Lemma 0.1.2. (Bezout's Lemma) Suppose $a, b \in \mathbb{Z} \setminus \{0\}$. Then there are $x, y \in \mathbb{Z}$ s.t. $ax + by = \gcd(a, b)$.

Proof. Let $a, b \in \mathbb{Z} \setminus \{0\}$. Consider $A = \{ax + by \mid x, y \in \mathbb{Z}\} \cap \mathbb{N}$

By the Well Ordering Principle since A has all non-negative terms it has a minimum lets call this d and call the associated $x, y \in \mathbb{Z}$, u and v .

By Theorem 0.1.1 $\exists q, r \in \mathbb{Z}$ s.t. $a = dq + r$ and $0 \leq r < d$

Case 1: $r = 0 \implies d \mid a$

Case 2: $0 < r < d \implies r = a - dq \implies r = a - (au + bv)q = a(1 - uq) + b(-vq) \in A$

$\implies r < d = \text{minimum} \implies \Leftarrow$

Thus $d \mid a$ and similarly $d \mid b$

Let $c \in \mathbb{N}$ s.t. $c \mid a$ and $c \mid b$

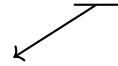
$\implies a = cm$ and $b = cn$ for $m, n \in \mathbb{Z}$

$d = au + bv = c(mu) + c(mv) = c(mu + mv) \implies c \mid d \implies d = \gcd(a, b)$ □

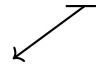
Algorithm 0.1.3. (Euclidean Algorithm)

Find $\gcd(272, 571)$ and $x, y \in \mathbb{Z}$ s.t. $272x + 571y = \gcd(272, 571)$

$$571 = 272 \cdot 2 + 27 \quad (1)$$



$$272 = 27 \cdot 10 + 2 \quad (2)$$



$$27 = 2 \cdot 13 + 1 \quad (3)$$

Last term before remainder is zero is gcd

$$\gcd(272, 571) = 1$$

$$2 = 272 - 27 \cdot 10$$

Use equation (2)

$$= 272(21) - 571(10)$$

$$1 = (571 - 272 \cdot 2) - 13(272(21) - 571)$$

Use equation (3) with (2)

$$= 571(131) + 272(-275)$$

1 Groups

1.1 Intro

Definition 1.1.1. A group is a set G together with a binary operation $*$: $G \times G \rightarrow G$ s.t.

- **(Closure)** For all $a, b \in G$ we have $a * b \in G$
- **(Identity)** There is $e \in G$ s.t. for all $a \in G$ $a * e = e * a = a$
- **(Inverses)** For all $a \in G$ there is an $a^{-1} \in G$ s.t. $a * a^{-1} = a^{-1} * a = e$
- **(Associativity)** For all $a, b, c \in G$ we have $a * (b * c) = (a * b) * c$

Definition 1.1.2. Given a group G and $g \in G$ we say the order of g is $n \in \mathbb{N}$ if n is the minimum value s.t. $g^n = g * g * \dots * g = e$
Write as: $\text{ord}(g) = |g| = n$

1.1.1 Exercises

Example 1.1.3. Let $G = \mathbb{R} \setminus \{-1\}$ and define $a * b = a + b + ab$. Show $(G, *)$ is an Abelian group

- (Abelian) For all $a, b \in G$ $a * b = a + b + ab = b + a + ba = b * a$
- (Closure) $\checkmark$
- (Identity) For all $a \in G$ $a * 0 = 0 * a = 0$
- (Inverses) For all $a \in G$ let $b = \frac{-a}{a+1}$.

$$\text{Thus, } a * b = b * a = a + b + ab = a + b(a+1) = \frac{-a}{a+1}(a+1) = a - a = 0$$

- (Associativity) For all $a, b, c \in G$

$$a * (b * c) = a * (b + c + bc) = a + b + c + bc + ab + ac + abc = (a + b + ab) * c = (a * b) * c$$

Therefore, G is an Abelian group

Example 1.1.4. Find all $x \in \mathbb{Z}$ s.t. $5x \equiv 4 \pmod{9}$

$$5x \equiv 4 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

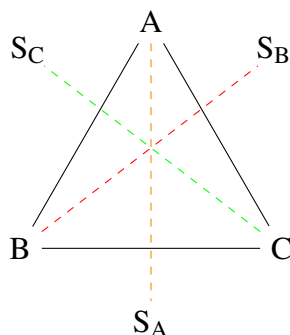
$$x = 9n + 8$$

$$2 * 5 = 10 \equiv 1 \pmod{9}$$

$$n \in \mathbb{Z}$$

1.2 Dihedral Groups

1.2.1 Rigid Motions of an Equilateral Triangle



Rotations

$r = 120^\circ$ rotation

$r^2 = 240^\circ$ rotation

$r^3 = 360^\circ$ rotation $= e = 0^\circ$ rotation

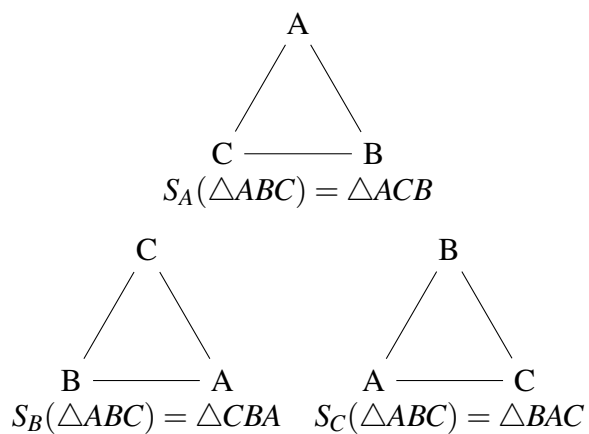
Reflections

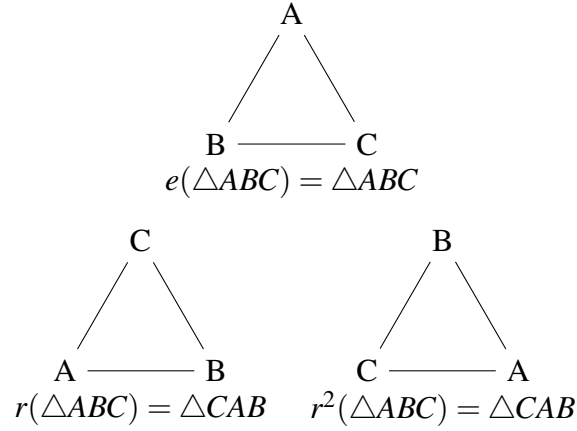
$$S_A : \triangle ABC \rightarrow \triangle ACB$$

$$S_B : \triangle ABC \rightarrow \triangle CBA$$

$$S_C : \triangle ABC \rightarrow \triangle BAC$$

Transformation Diagrams





Operations

$$\begin{aligned}
 r \circ S_A(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S_C(\triangle ABC) & r \circ S_A &= S_C \\
 r^2 \circ S_A(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S_B(\triangle ABC) & r^2 \circ S_A &= S_B
 \end{aligned}$$

Now we only need S_A so lets call this S

Thus we only need S, r, r^2, e

Also $S^2 = r^3 = e$

How S, r, r^2 Commute

$$\begin{aligned}
 r \circ S(\triangle ABC) &= r(\triangle ACB) = \triangle BAC = S \circ r^2 \\
 r^2 \circ S(\triangle ABC) &= r^2(\triangle ACB) = \triangle CBA = S \circ r
 \end{aligned}$$

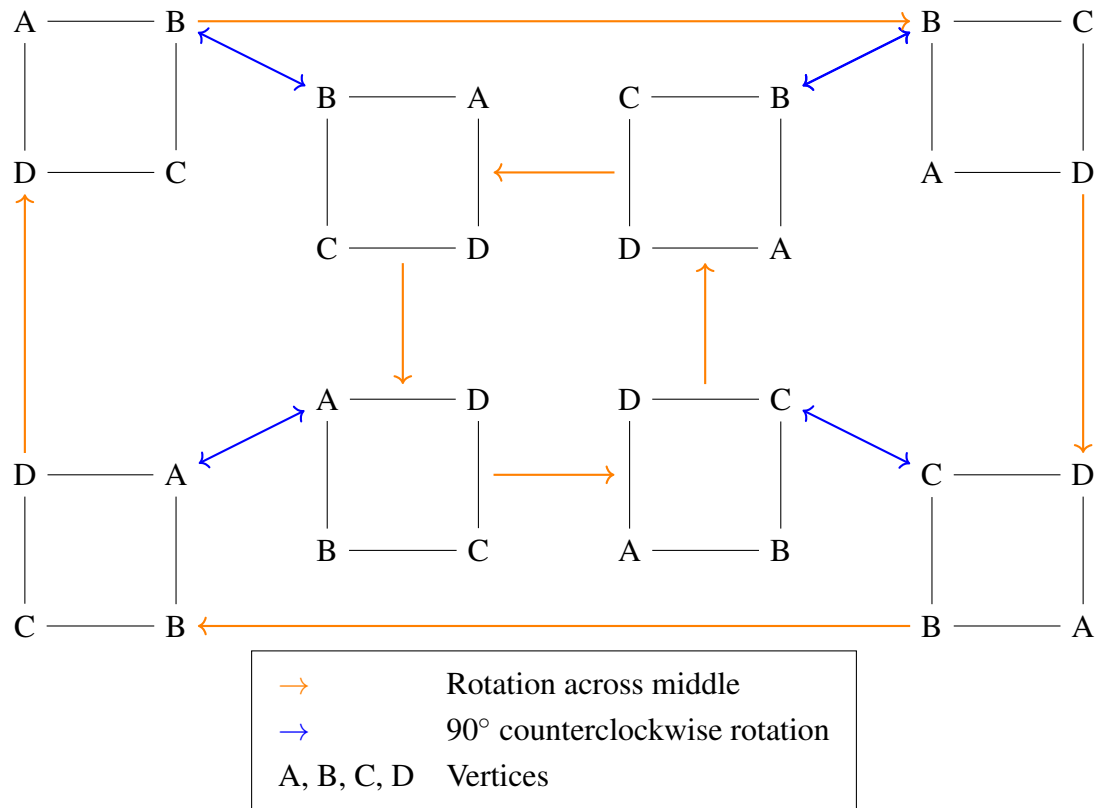
Definition 1.2.1. The group of symmetries of an equilateral triangle is $D_3 = \{e, r, r^2, S, Sr, Sr^2\}$, where $r^3 = S^2 = e$, $rS = Sr^2$

Dihedral group of order $2 \cdot 3 = 6$

Compact Notation $D_3 = \langle r, s^1 \mid r^3 = S^2 = e, rS = Sr^2 \rangle$

¹The generators of D_3 meaning you can create all other operations from them

1.2.2 Symmetry Group for a Square



Rotations

$r = 90^\circ$ rotation
 $r^2 = 180^\circ$ rotation
 $r^3 = 270^\circ$ rotation
 $r^4 = 360^\circ$ rotation = $e = 0^\circ$ rotation

Reflections

$$s : \square ABCD \rightarrow \square BCDA$$

How does s, r Commute

From the diagram we see that $sr = r^3s \implies rs = sr^3$

Definition 1.2.2. The group of symmetries of a square is $D_4 = \{e, r, r^2, r^3, s, sr, sr^2, sr^3\}$

Dihedral group of order $2 \cdot 4 = 8$

$$D_4 = \langle s, r \mid r^4 = s^2 = e, sr = r^3s \rangle$$

1.2.3 Symmetry Group of regular n-gon

Rotation: $r = \frac{360^\circ}{n} \circlearrowleft (e, r, r^2, \dots, r^{n-1})$

Reflection: $s = \text{reflection across middle } (s, sr, sr^2, \dots, sr^{n-1})$

Claim: $rs = sr^{n-1}$.

Proof. We only need 3 points since our shape is defined by the order of those 3 points

$$r \circ s(n, 1, 2) = r(2, 1, n) = (1, n, n-1)$$

$$s \circ r^{n-1}(n, 1, 2) = s(n-1, n, 1) = (1, n, n-1)$$

Thus $rs = sr^{n-1}$ □

Definition 1.2.3. The dihedral group D_n is the symmetry group of a regular n-gon.

$$D_n = \langle s, r \mid r^n = s^2 = e, rs = sr^{n-1} \rangle$$

Claim: For all $1 \leq m \leq n-1$ we have $r^m s = sr^{n-m}$.

Proof. Base Case: $m = 1 \implies rs = sr^{n-1}$ ✓

Inductive Step: Assume for $k \geq 1$ the statement is true

Consider:

$$r^{k+1}s = rr^k s$$

$$= rsr^{n-k}$$

$$= sr^{n-1}r^{n-k}$$

$$= sr^n r^{n-(k+1)}$$

$$= sr^{n-(k+1)}$$

Apply inductive hypothesis on $r^k s$

□

1.2.4 Exercises

Example 1.2.4. Simplify

- $(sr^5)r(sr^3)$ in D_4

$$= (sr)r(rs)$$

$$= e$$

- $(sr^4)(sr^2)$ in D_6

$$= (r^2s)(sr^2) \quad \text{Use } r^m s = sr^{n-m}$$

$$= r^4$$

- $r^3(sr)$ in D_4

$$= r^3(r^3s)$$

$$= r^2s$$

Example 1.2.5. The Cayley table of the D_3 can be written

	e	r	r^2	s	rs	r^2s
e	e	r	r^2	s	rs	r^2s
r	r	r^2	e	rs	r^2s	s
r^2	r^2	e	r	r^2s	s	rs
s	s	r^2s	rs	e	r^2	r
rs	rs	s	r^2s	r	e	r^2
r^2s	r^2s	rs	s	r^2	r	e

2 Symmetric Groups

Definition 2.0.1. Given a set X , a permutation of X is a bijection $X \rightarrow X$
 $S_X = \{ \sigma : X \rightarrow X \mid \sigma \text{ is a bijection} \}$

Lemma 2.0.2. For all sets $X \neq \emptyset$, S_X is a group

Proof.

- Closure: Suppose $\sigma, \mu \in S_X$
 Consider $\sigma \circ \mu$ since it is a composition of bijections it is also a bijection.
 Also functions go from $X \rightarrow X$ thus $\sigma \circ \mu$ is the same.
 Therefore, the composition is a bijection that is mapped from $X \rightarrow X$ thus $\sigma \circ \mu \in S_X$
- Identity: Consider $e : X \rightarrow X$ s.t. $e(x) \mapsto x$
 Thus $e \in S_X$ and for all $\sigma \in S_X$ $\sigma \circ e(x) = \sigma(x)$
 Thus $\sigma \circ e = \sigma$. Also, it follows that $e \circ \sigma = \sigma$
- Inverses: Since S_X has only bijective functions all bijective functions have inverses.
- Associativity: We have associativity by the fact that function composition has associativity.

□

Definition 2.0.3. The permutation group on n -objects S_n is the S_X with $X = [n]$
 Observe $|S_n| = n!$

Example 2.0.4. $\sigma, \mu \in S_4$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$$

$$\sigma^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \mu^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$$

$$\sigma\mu = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix} \mu\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 2 & 1 & 3 \end{pmatrix}$$

2.1 Cycle Notation

Example 2.1.1. $\sigma, \mu \in S_5$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix} \mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 2 & 1 & 3 \end{pmatrix}$$

$$\sigma = (13245) \mu = (14)(253)$$

Example Calculations:

$$\begin{aligned} \sigma\mu &= (13245)(14)(253) \\ &= (152)(34) \\ \mu\sigma &= (14)(253)(13245) \\ &= (12)(354) \\ \mu^2 &= (14)(253)(14)(253) \\ &= (14)^2(253)^2 = (235) \end{aligned}$$

Lemma 2.1.2. *Disjoint cycles are commutative*

Proof. Let α, β be disjoint cycles of $[n]$

Let $\alpha = (a_1, a_2, \dots, a_k)$, and $\beta = (b_1, b_2, \dots, b_j)$

Let $x \in [n]$

Consider: $\alpha\beta(x), \beta\alpha(x)$

If $x \in \alpha$ then $x \notin \beta \implies \beta(x) = x \implies \alpha\beta(x) = \alpha(x) = a_i$
 $\implies \beta\alpha(x) = \beta(a_i) = a_i$ since $a_i \notin \beta$

Use similar reasoning for if $x \in \beta$

$\implies \alpha\beta = \beta\alpha$

□

Lemma 2.1.3. *Every $\sigma \in S_n$ can be written as disjoint cycles*

Lemma 2.1.4. *The order of an m -cycle is m*

Lemma 2.1.5. Suppose G is a group and $\text{ord}(g) = n$. Then $g^m = e \iff n \mid m$

Proof. Let G be a group and $\text{ord}(g) = n$

($\Leftarrow$)

Assume $n \mid m$

$$\implies m = nk, k \in \mathbb{Z}$$

$$g^m = g^n k = (g^n)^k = (e)^k = e$$

($\Rightarrow$)

Assume $g^m = e$ and use division algorithm to write $m = nq + r$ s.t. $0 \leq r < n$

$$e = g^m = g^{nq+r} = (g^n)^q g^r = g^r \implies r = 0 \implies n \mid m$$

□

Theorem 2.1.6. If $\sigma_1, \sigma_2, \dots, \sigma_k$ are disjoint cycles of length $a_1, a_2, \dots, a_k$ and $\sigma = \sigma_1 \sigma_2 \dots \sigma_k$. Then $\text{ord}(\sigma) = \text{lcm}(a_1, a_2, \dots, a_k)$.

Proof. Let $m = \text{ord}(\sigma)$, $l = \text{lcm}(a_1, a_2, \dots, a_k)$

Since l is a lcm it is also a multiple so $l = a_j q_j$ for $1 \leq j \leq k$ where $q_j \in \mathbb{Z}$

$$\sigma^l = \sigma_1^l \sigma_2^l \dots \sigma_k^l$$

We can do this because of Lemma 2.1.6

$$= (\sigma_1^{a_1})^{q_1} \dots (\sigma_k^{a_k})^{q_k}$$

$$= (1)$$

Apply Lemma 2.1.8

$$\implies m \mid l$$

Also, $(1) = \sigma^m = \sigma_1^m \dots \sigma_k^m$ since they are disjoint none of them are inverses of each other

$$\implies \sigma_1^m = \dots = \sigma_k^m = (1)$$

(Lemma 2.1.9) $\implies a_1 \mid m \dots a_k \mid m \implies l \mid m$

Thus, since $l \mid m$ and $m \mid l$ $m = l$

□

Example 2.1.7.

- Find an element of order 12 in S_7

$(1234)(567) \in S_7$ by Theorem 2.0.10 $\implies$ the order is $3 \cdot 4 = 12 = \text{lcm}(3,4)$

- Possible orders of elements in S_5

- 1: (1) , $5=1+1+1+1+1$
- 2: (12) , $5=2+1+1+1+1$ or $(12)(34)$, $5=2+2+1$
- 3: (123) , $5=3+1+1$
- 4: (1234) , $5=4+1$
- 5: (12345) , $5=5$
- 6: $(12)(345)$, $5=2+3$